CAET ADVANCED CERTIFICATION · STUDENT HANDBOOK

Section 1: Regulatory Compliance & Approved Data

Aircraft Electronics Association · Workforce Development Division

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Why This Matters

Every federal aviation regulation in Title 14 of the Code of Federal Regulations is written in response to something that killed people. Part 43 was not drafted by bureaucrats who wanted to make paperwork; it was drafted after decades of accident investigation showing that aircraft built without controlled data, repaired without approved methods, and released to service without accountable signatures kill passengers at rates the public refuses to tolerate. When you sign a logbook entry under 14 CFR 43.9 or return an aircraft to service under 43.7, you are not filling out a form. You are invoking the full force of federal law to certify that your work meets the performance rules of 14 CFR 43.13 and was accomplished with approved data. That signature is the contract between you and the flight crew, the passengers, and the people on the ground beneath the flight path.

The CAET technician who installs a new GPS navigator without an STC, who substitutes non-PMA parts to meet a deadline, or who signs a 337 without actually inspecting the work has not committed a paperwork error. They have committed a federal offense under 14 CFR 43.12 (falsification of records) and 49 USC 44709 (certificate actions). The FAA's enforcement history is filled with cases where a single falsified logbook entry or a single unapproved repair led to certificate revocation, civil penalties exceeding $100,000, and in fatal cases, criminal prosecution under 18 USC 38 (fraud involving aircraft parts). Revocation under 14 CFR 65.20 is not a suspension you wait out. It is permanent. A technician whose A&P is revoked for falsification will never hold another mechanic certificate, and most Part 145 repair stations are barred by their ops specs from employing anyone who has been denied or revoked under 65.12.

The regulatory framework is not an obstacle to your work; it is the framework that makes your work possible. 14 CFR 65.81 gives you, the AMT, the privilege to perform maintenance and approve aircraft for return to service. 14 CFR 145.51 gives your shop the authority to do the same at the organizational level. 14 CFR 91.403 places the owner at the top of the airworthiness chain, and 43.5 and 43.9 create the audit trail that proves compliance. Master these regulations and you are a trusted professional whose signature carries weight across the entire industry. Misunderstand them and you are one FSDO audit away from losing everything. This module exists to make sure you are the first kind of technician.

FROM THE PQS SOURCE

This section tests what separates a parts-installer from a professional avionics technician: knowing the rules that govern every piece of work you do. In a Part 145 shop, every installation, every repair, and every inspection must be backed by approved data.

Performing work without it is not just bad practice—it is a violation of federal aviation regulations that can ground an aircraft, shut down a shop, and end a career.

Your evaluator will sign off these tasks as you demonstrate them in your daily work. The oral board will probe whether you truly understand the regulatory framework or just follow shop procedures without knowing why.

Key Terms & Definitions

Master this vocabulary before continuing. The evaluator expects you to use these terms correctly in the oral board.

Repair Station (Part 145)

An organizational certificate issued under 14 CFR Part 145 that authorizes a facility to perform maintenance, preventive maintenance, and alterations on aircraft, engines, propellers, appliances, or components. Ratings are defined in 14 CFR 145.59 (Airframe, Powerplant, Radio, Instrument, Accessory, Limited) and the specific work authority is spelled out in the Operations Specifications (OpSpecs).

AMT / A&P Mechanic

An Aviation Maintenance Technician holding a mechanic certificate under 14 CFR Part 65 Subpart D. The Airframe and Powerplant (A&P) rating is a combination of two individual ratings issued under 14 CFR 65.77. Privileges are defined in 14 CFR 65.81 (general) and 65.85/65.87 (airframe/powerplant specifically).

Repairman Certificate

A certificate issued under 14 CFR 65.101 that authorizes an individual to perform specific maintenance tasks for a specific employer (typically a Part 145 station or Part 121/135 operator). Unlike an A&P, a repairman certificate is not portable — it is valid only for the named employer and the specific work listed on the certificate under 14 CFR 65.103.

14 CFR Part 43 (Maintenance)

The regulation governing maintenance, preventive maintenance, rebuilding, and alteration of US-registered aircraft. Key sections: 43.3 (authorized persons), 43.5 (required recordkeeping to release to service), 43.7 (approval for return to service), 43.9 (content, form, and disposition of records), 43.11 (annual/100-hour content), 43.13 (performance rules — methods/techniques/practices), 43.15 (additional performance rules), Appendix A (major alterations/repairs), Appendix B (Form 337 and recording of major repairs/alterations), Appendix D (annual/100-hour inspection scope).

14 CFR Part 65 (Airmen Other Than Flight Crew)

Governs certification of mechanics, repairmen, parachute riggers, dispatchers, and inspection authorization holders. Subpart D covers mechanics (65.71-65.95); Subpart E covers repairmen (65.101-65.104); Subpart G covers inspection authorization (65.91-65.95). Key enforcement sections: 65.12 (offenses involving controlled substances), 65.15 (duration of certificates), 65.20 (making or causing false records).

14 CFR Part 91 (General Operating Rules)

The general operating and flight rules applicable to all civil aircraft in the United States. Maintenance-relevant sections: 91.403 (owner/operator responsibility for airworthiness), 91.405 (maintenance required), 91.407 (operation after maintenance/repair/alteration), 91.409 (inspections — annual, 100-hour, progressive, AAIP), 91.411/91.413 (altimeter/transponder/static system checks), 91.417 (maintenance records retention), 91.421 (rebuilt engine records).

14 CFR Part 145 (Repair Stations)

The operating rule for certificated repair stations. Key sections: 145.51 (application for certificate), 145.53 (issue of certificate), 145.59 (ratings), 145.109 (required equipment/materials/data), 145.201 (privileges and limitations), 145.211 (quality control system), 145.217 (contract maintenance), 145.221 (service difficulty reports).

Type Certificate (TC)

The foundational FAA approval document for a specific aircraft, engine, or propeller design issued under 14 CFR Part 21 Subpart B. The TC Data Sheet (TCDS) defines the certification basis, limitations, and required equipment. Any modification that departs from the TC configuration is — by definition — an alteration to type design.

Supplemental Type Certificate (STC)

An FAA approval issued under 14 CFR Part 21 Subpart E that authorizes a major change to the type design of a previously type-certificated product. The STC consists of an approval letter, an Approved Model List (AML) of applicable aircraft, installation instructions, and Instructions for Continued Airworthiness (ICA) per 14 CFR 21.50. Legally, the STC package is Approved Data when used on an eligible airframe and installed per its instructions.

Parts Manufacturer Approval (PMA)

An FAA approval under 14 CFR 21.303 authorizing a manufacturer to produce replacement or modification parts for installation on type-certificated products. A PMA part must be marked per 14 CFR 45.15 and is accompanied by a Statement of Conformity (8130-3) or a PMA plate. Non-PMA parts for flight-critical systems are not approved for installation regardless of physical similarity.

Approved Data

Technical information that the FAA has formally approved as a basis for a major repair or major alteration. Sources include: the TCDS, STC data, DER-approved 8110-3 data, manufacturer's ICA (when FAA-accepted as part of the TC), Airworthiness Directive

documents, AC 43.13-1B/-2B when used for minor alterations or approved by FSDO for field approval, and data approved by a Designated Organizational Authority (DOA) under 14 CFR 183.

AC 43.13-1B / AC 43.13-2B

FAA Advisory Circulars titled 'Acceptable Methods, Techniques, and Practices — Aircraft Inspection and Repair' (1B) and 'Acceptable Methods, Techniques, and Practices — Aircraft Alterations' (2B). Acceptable for minor alterations/repairs only; for major alterations, AC 43.13 can serve as Approved Data only when a FSDO inspector signs Block 3 of Form 337 for field approval, per FAA Order 8300.10.

Major vs. Minor Alteration

Defined in 14 CFR 1.1 and 14 CFR 43 Appendix A(a). A major alteration 'might appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other qualities affecting airworthiness' or 'is not done according to accepted practices.' All others are minor. Major alterations require Approved Data and FAA Form 337; minor alterations require only a logbook entry per 43.9.

Major vs. Minor Repair

Defined in 14 CFR 1.1 and 14 CFR 43 Appendix A(b). A major repair is one that, if improperly done, might appreciably affect airworthiness, or is not done by elementary operations. Examples in Appendix A include repairs to ribs, spars, control surfaces, pressure vessels, and flight control systems. Major repairs require Form 337; minor repairs require a 43.9 logbook entry.

FAA Form 337 (Major Repair and Alteration)

The official record of a major repair or major alteration, required by 14 CFR 43 Appendix B. Block 8 describes the work; Block 3 is signed by the person approving the data; Block 6 is signed by the person approving the aircraft for return to service. One copy goes to the owner (kept with aircraft records per 91.417), one copy must be submitted to the FAA Aircraft Registration Branch within 48 hours of aircraft return to service, per 43 Appendix B(a)(3).

8110-3 DER Data

FAA Form 8110-3, Statement of Compliance With Airworthiness Standards, signed by a Designated Engineering Representative (DER) under authority of 14 CFR 183.29. DER-approved data can be used as Approved Data for a specific major repair or alteration when the DER's delegation matches the work. 8110-3 is most commonly used for one-off repairs/alterations where an STC is not economically justified.

Return to Service (RTS)

The legal act of certifying that an aircraft or component is airworthy after maintenance, performed under 14 CFR 43.7. For a major repair or alteration, RTS signature goes in Block 7 of Form 337. For routine maintenance, it is the 43.9 logbook entry. The RTS signer must hold

appropriate certificate and authority (A&P for routine; IA for annuals and major RTS outside a 145 station; repair station certificate-holder for work under 145).

MEL (Minimum Equipment List)

An operator-specific list of inoperative equipment that the aircraft may be dispatched with, under 14 CFR 91.213(a). An MEL must be approved by the operator's FSDO via a Letter of Authorization (LOA) under Part 91, or as part of the OpSpecs under Part 121/135. An MEL is always more restrictive than the MMEL on which it is based.

MMEL (Master Minimum Equipment List)

An FAA-issued document (by aircraft type) that establishes the baseline from which an operator's MEL is developed, issued by the Flight Standardization Board per FAA Order 8900.1. The MMEL is not directly usable by an operator — it must be tailored into an MEL. Part 91 operators of turbine aircraft may use the MMEL directly only if they have a Letter of Authorization permitting it.

CDL (Configuration Deviation List)

A list of non-structural external parts (inspection panels, fairings, access covers) that may be missing for a flight, published as an appendix to the AFM for some type certificates. Unlike MEL items, CDL deviations do not require a maintenance deferral entry because they are not inoperative equipment — they are missing secondary structure addressed in the TC.

Airworthiness Directive (AD)

A legally binding notice of unsafe condition issued under 14 CFR Part 39. ADs are mandatory — compliance is required by 39.7, and operation in violation of an AD is a 91.403 violation. ADs are the only FAA publication that has the force of regulation without requiring additional rulemaking; they apply to the registered owner regardless of aircraft location.

PART II — KNOWLEDGE REQUIREMENTS

1.2.1 — 14 CFR Part 43: Who Can Perform Maintenance

14 CFR Part 43 is the legal foundation for all maintenance on US-registered civil aircraft (except Part 121/135 aircraft, which operate under additional rules). Part 43 defines what maintenance is (43.1-43.2), who may perform it (43.3), what records are required (43.9, 43.11), and how the work must be performed (43.13). For the avionics technician working inside a Part 145 station, your personal authority is extended through the repair station's certificate — but you remain accountable to Part 43's performance rules as an individual.

FROM THE PQS SOURCE

Part 43 is the backbone of aircraft maintenance regulation. It defines what constitutes maintenance, who is authorized to perform it, and what documentation is required. For avionics work, the key distinctions are: As an avionics technician working under a Part 145 certificate, your authority to perform maintenance comes from the repair station's ratings, not from your personal certificates. The repair station Operations Specifications define exactly what work the shop is authorized to do. Working outside those ops specs is a certificate violation.

KEY POINTS

- 14 CFR 43.3(a)-(d) enumerates who may perform maintenance: certificated mechanic (A&P), repairman, holder of an air carrier operating certificate, certificated repair station, pilot (limited preventive maintenance per 43.3(g) and Appendix A(c)), and manufacturer (43.3(j)).
- 14 CFR 43.3(d) specifically authorizes a certificated repair station to perform maintenance according to Part 145 — this is how a Part 145 station derives its work authority, and it is why your shop's OpSpecs are legally binding on you when you work there.
- 14 CFR 43.7(a)-(f) defines who may approve an aircraft for return to service. For major repairs and major alterations, only an A&P with IA (65.95), a repair station (145.201(c)), or the manufacturer may approve RTS. An A&P without IA can return to service after routine maintenance but cannot sign off on a major alteration.
- 14 CFR 43.13(a) requires that 'each person performing maintenance, alteration, or preventive maintenance on an aircraft, engine, propeller, or appliance shall use the methods, techniques, and practices prescribed in the current manufacturer's maintenance manual or Instructions for Continued Airworthiness prepared by its manufacturer, or other methods, techniques, and practices acceptable to the Administrator.'
- 14 CFR 43.13(b) requires you to perform work 'in such a manner and use materials of such a quality, that the condition of the aircraft, airframe, aircraft engine, propeller, or appliance worked on will be at least equal to its original or properly altered condition.' This is the 'original or better' standard that underlies every repair.
- 14 CFR 43.12 makes it a federal offense to make a fraudulent or intentionally false entry in any maintenance record. Violations are prosecuted under 49 USC 44709 (certificate actions) and, in cases involving flight-critical systems, under 18 USC 38 (fraud involving aircraft parts), which carries penalties up to 15 years imprisonment and fines exceeding $500,000.
- Working outside your repair station's OpSpecs is treated under 14 CFR 145.201(c) as unauthorized maintenance and subjects both the station and the individual technician to

certificate action.

1.2.2 — 14 CFR 91.403: Owner/Operator Responsibility

14 CFR 91.403(a) places the owner or operator at the apex of the airworthiness chain: 'The owner or operator of an aircraft is primarily responsible for maintaining that aircraft in an airworthy condition, including compliance with Part 39 (Airworthiness Directives).' Technicians enable compliance; owners are ultimately accountable. This distinction shapes how you document work and how you handle customer authorization.

FROM THE PQS SOURCE

Here is the concept that surprises many technicians: the owner/operator is ultimately responsible for the airworthiness of the aircraft—not the mechanic, not the repair station. Section 91.403(a) states that the owner or operator is primarily responsible for maintaining the aircraft in an airworthy condition. When you perform maintenance, you are acting on behalf of the owner. This is why proper work orders, customer authorization, and return-to-service documentation matter—they define the scope of what the owner authorized you to do.

KEY POINTS

- 91.403(a): The owner/operator is primarily responsible for airworthiness, including AD compliance under Part 39.
- 91.403(b): No person may perform maintenance on a civil aircraft other than as prescribed in Part 43 and other applicable regulations — this is the hook that applies Part 43 to all civil aircraft maintenance.
- 91.403(c): No person may operate an aircraft for which a manufacturer's maintenance manual or ICA has been issued that contains an Airworthiness Limitations Section unless the mandatory replacement times, inspection intervals, and related procedures have been complied with.
- 91.405 requires the owner/operator to have inspections performed in accordance with Part 91 Subpart E and to ensure maintenance personnel make appropriate entries in the aircraft's maintenance records.
- 91.407 prohibits operation of an aircraft after maintenance, preventive maintenance, rebuilding, or alteration until it has been approved for return to service and the maintenance record entry required by 43.9 (or 43.11 for inspections) has been made. After a major alteration that may have appreciably changed flight characteristics, 91.407(b) requires a pilot who holds at least a private certificate and is rated for the

aircraft to perform a flight test and log the results before the aircraft can carry passengers.

- 91.409 defines the inspection framework: annual for non-commercial aircraft under 91.409(a)(1); 100-hour for aircraft used for hire under 91.409(b); progressive under 91.409(d); and AAIP (Approved Aircraft Inspection Program) under 91.409(e)-(f) for large/turbine aircraft.
- 91.417 governs maintenance records retention. Per 91.417(b)(1), 43.9 records of routine maintenance are retained for 1 year or until superseded. Per 91.417(b)(2), 43.11 inspection records, total time in service, status of life-limited parts, time since last overhaul of required-overhaul items, airworthiness status, AD compliance, and copies of all Form 337s are retained for the life of the aircraft and transferred with the aircraft upon sale.

1.2.3 — Approved Data Hierarchy

Not all technical documents carry the same regulatory weight. When you plan a major repair or major alteration, you must select Approved Data from a specific, legally defined hierarchy. Using the wrong data source — even if technically correct — is a violation of 14 CFR 43.13 and a ground for enforcement action.

FROM THE PQS SOURCE

Not all technical data carries the same authority. When planning avionics work, you must use data from the highest applicable level:

KEY POINTS

- Tier 1 (Type Design Data): The Type Certificate Data Sheet (TCDS), the manufacturer's maintenance manual, the component maintenance manual (CMM), and the Instructions for Continued Airworthiness (ICA) required by 14 CFR 21.50. When the manufacturer's data covers your work, it is always the first source.
- Tier 2 (STC Data): A Supplemental Type Certificate package, including the STC Approval Letter, the Approved Model List (AML), the installation drawings, the master drawing list, and the STC's ICA. STC data is Approved Data only for aircraft on the AML and only when installed per the instructions — deviation invalidates the approval.
- Tier 3 (DER-Approved Data): FAA Form 8110-3 signed by a Designated Engineering Representative whose delegation matches the work. Per 14 CFR 183.29 and FAA Order 8110.37, DER 8110-3 data is binding for the specific aircraft and modification described — it is not a generic approval.

- Tier 4 (FSDO Field Approval): FAA Form 337 with Block 3 signed by an FAA Airworthiness Safety Inspector approving AC 43.13-1B/2B data or other acceptable data as Approved Data for a specific one-off major repair or alteration, per FAA Order 8300.10, Volume 3, Chapter 1.
- Tier 5 (DOA / ODA Data): Designated Organizational Authority (DOA) or Organization Designation Authorization (ODA) data approved under 14 CFR Part 183 Subpart D. Applies when an OEM or operator has been delegated organizational approval authority.
- Acceptable Data vs. Approved Data: AC 43.13-1B is Acceptable Data for minor repairs/alterations (you can use it without additional FAA signature). It becomes Approved Data for majors only when a FSDO inspector field-approves it on Block 3 of a 337.
- Airworthiness Directives are Approved Data for the specific corrective action they require — per 14 CFR 39.7, you must comply using the methods specified in the AD or FAA-approved alternative methods of compliance (AMOCs).

1.2.4 — STC vs. AC 43.13-1B: When Each Applies

Deciding whether AC 43.13-1B is sufficient or whether an STC is required depends on whether the work is a major alteration (per 14 CFR 1.1 and Part 43 Appendix A) and whether it changes the aircraft's type design. The classification drives the data source, the documentation (337 vs. 43.9), and the approval authority. Getting this wrong is one of the most common and most consequential errors an avionics shop can make.

FROM THE PQS SOURCE

The decision tree is straightforward: Is the work a repair? AC 43.13-1B methods are generally acceptable for repairs to existing installations. Replacing a damaged antenna with the same type, repairing a chafed wire, replacing a connector—these are repairs. Is the work a minor alteration? AC 43.13-1B may apply for minor alterations that don’t change the aircraft’s type design. Adding a reading light, changing a USB charging port, installing a simple audio jack. Is the work a major alteration? An STC or field approval (Form 337 with FSDO approval) is required. Installing a new GPS navigator, adding an autopilot, upgrading from analog to glass cockpit—these change the aircraft’s type design and require STC-level approval.

KEY POINTS

- Repair with like-for-like replacement (e.g., swapping a failed comm radio with an identical unit per the wire list): AC 43.13-1B methods are acceptable; work is recorded as a 43.9 logbook entry. No 337 required because the type design is not changed.

- Minor alteration not changing type design (e.g., adding a USB charging port powered by an existing accessory bus, a reading light, a simple audio jack splitter): AC 43.13-1B/2B methods are acceptable as Acceptable Data for minor alterations. Logbook entry per 43.9 — no 337.
- Major alteration (e.g., installing a new IFR GPS navigator, adding a digital autopilot, upgrading from analog CDI to an EFIS): STC or field-approved 337 required. This changes type design and falls under Part 43 Appendix A(a) criteria (appreciably affects performance or flight characteristics). Always a 337.
- Major repair (e.g., repair of a pressure vessel, repair of a control surface, splicing a flight-critical wire bundle where AC 43.13-1B methods alone are insufficient): Must use Approved Data — manufacturer's CMM/ICA, STC/DER data, or field-approved 337. Always a 337.
- AC 43.13-1B Paragraph 1-3 explicitly states the circular 'may be used as approved data for major alterations when the user has determined that it is appropriate to the product being altered, it is directly applicable to the alteration being made, and it is not contrary to data provided by the manufacturer of the product or contrary to data developed by the holder of a Supplemental Type Certificate.' This is only operative when a FSDO inspector field-approves on a 337.
- Audit your decision: If the installation requires new antennas drilled through skin, new circuit protection, weight-and-balance changes, interface with flight instruments, or power draw from a protected bus — you are almost certainly in major alteration territory.
- Common trap: A manufacturer's install manual alone does not authorize installation. If the installation is a major alteration, you need an STC or a field-approved 337 even though the manufacturer's book describes the installation steps perfectly.

1.2.5 — Part 145 Repair Station Privileges

14 CFR Part 145 is the regulatory framework that governs your shop's legal identity as a certified repair station. Your personal authority to perform maintenance inside the shop flows from the repair station certificate — not from your A&P certificate alone. Understanding the shop's OpSpecs, ratings, and quality control system is essential to knowing exactly what you can and cannot sign off on.

FROM THE PQS SOURCE

A Part 145 repair station certificate authorizes the station to perform maintenance, preventive maintenance, and alterations on articles (aircraft, engines, propellers, appliances, or components) as specified in its Operations Specifications. The key limitations to understand: The station can only perform work within its ratings (limited, airframe,

powerplant, radio, instrument, accessory) Work must be performed according to the station’s Repair Station Manual (RSM) and Quality Control Manual (QCM) Technicians working under the station’s certificate do not need individual A&P certificates, but they must be trained and authorized per the station’s training program The station’s Accountable Manager is responsible for compliance with all regulations Contract maintenance must follow specific rules under 145.217

KEY POINTS

- 14 CFR 145.51 governs certificate application; 145.53 governs issuance. The shop's certificate is accompanied by Operations Specifications (OpSpecs) that spell out the specific articles, ratings, and locations the shop is authorized to work on.
- 14 CFR 145.59 defines six ratings: Airframe (class 1-4), Powerplant (class 1-3), Propeller (class 1-2), Radio (class 1-3), Instrument (class 1-4), Accessory (class 1-3). A shop's Radio class 1 covers communications/navigation/pulse equipment; class 2 covers radar; class 3 covers radio navigation aids including ILS/VOR/DME/GPS.
- 14 CFR 145.109 requires the repair station to have the equipment, tools, materials, and data necessary to perform the ratings issued. This is why your shop maintains current STC libraries, manufacturer technical publications, AC 43.13-1B, and calibrated test equipment.
- 14 CFR 145.201(a) grants privileges: perform maintenance, preventive maintenance, and alterations per the station's ratings and OpSpecs; approve for return to service per 43.7(c).
- 14 CFR 145.201(c) imposes limitations: the station may not perform maintenance on an article for which it is not rated, and it may not approve for return to service any article after major repair/alteration if the work was not performed in accordance with approved/acceptable data applicable to the article.
- 14 CFR 145.161 requires each repair station to have a Repair Station Manual (RSM) that describes how maintenance is performed, and a Quality Control Manual (QCM) that describes how work is inspected. The RSM/QCM is enforceable — FAA audits cite shops for failing to follow their own manuals.
- 14 CFR 145.163 requires that each technician working under the station's authority be trained per the station's Training Program (required by 145.163(a)), and the station must maintain training records. This is how a non-certificated technician gains authority to work on your aircraft — the RS training program substitutes for individual certification.
- 14 CFR 145.217 governs contract maintenance: the station can contract work to another certificated source, but the contracting station retains responsibility under its certificate and must list the contract maintenance provider on its OpSpecs.

1.2.6 — Airworthiness Directives

Airworthiness Directives are the FAA's mechanism for mandating correction of unsafe conditions identified after type certification. AD compliance is not optional, not a 'best practice,' and not a customer service preference — it is federal law under 14 CFR Part 39. Every installation and every annual requires an AD check, and failure to identify and comply with an applicable AD is a direct 91.403 violation against the owner and a competence-and-diligence issue against the technician.

FROM THE PQS SOURCE

ADs are mandatory safety corrections. Before any avionics installation, you must check whether any ADs apply to the equipment being installed, the equipment being removed, or the aircraft systems affected by the installation. AD research is typically done through the FAA AD database (rgl.faa.gov), manufacturer service information, and type-club resources. Key point: Missing an applicable AD during an installation is a serious regulatory violation. The AD check must be documented in your work package, even if no ADs apply—record that you checked and found none applicable. The oral board will test whether you understand the consequences of regulatory non-compliance, not just the rules themselves. Be prepared to explain what happens when things go wrong: Your evaluator will sign off each task as you demonstrate it in your normal work. Here is what each task requires and how to prepare.

KEY POINTS

- 14 CFR 39.3: 'The FAA's airworthiness directives are legally enforceable rules that apply to the following products: aircraft, aircraft engines, propellers, and appliances.' ADs apply to products regardless of where they are installed or who owns them.
- 14 CFR 39.7: 'Anyone who operates a product that does not meet the requirements of an applicable airworthiness directive is in violation of this section.' Compliance is the operator's legal duty under 91.403(a); enabling compliance is the technician's professional duty.
- 14 CFR 39.19 allows an Alternative Method of Compliance (AMOC) to be approved by the FAA when the operator demonstrates equivalent level of safety. AMOCs must be formally approved — you cannot unilaterally substitute a different procedure.
- AD research sources: FAA Dynamic Regulatory System (drs.faa.gov) is the current authoritative source (having replaced rgl.faa.gov). Search by aircraft make/model, engine, propeller, and appliance. Always document the search date, database queried, search terms, and results.
- Emergency ADs and Airworthiness Concern Sheets (ACS): Emergency ADs can be issued without the standard NPRM process under 14 CFR 39.17 when an unsafe condition exists

that requires immediate action. Operators are typically notified directly; compliance dates can be as short as 'before next flight.'

- AD types: One-time inspection/action, recurring inspection (must comply initially and at repeating intervals), terminating action (a modification that ends the recurring requirement), and superseded ADs (always check if the original AD has been superseded — always comply with the current AD).
- Required records: Per 14 CFR 91.417(a)(2)(v), the owner must maintain a current status of applicable ADs including the method of compliance, the AD number and revision date, and for recurring ADs, the time/date next action is required. Your logbook entry must cite the AD number, compliance method, and recurrence interval.
- AD documentation in installation work: Every work package must include an AD search result. Even if no ADs apply, record that you checked the date, the database, the part numbers searched, and 'no applicable ADs found.' This proves diligence and satisfies 91.417 recordkeeping.

PART III — TASK PREPARATION

Task 1.4.1 — Determine Required Approved Data

How to demonstrate: When your evaluator presents a scenario (or when one arises in your work), walk through the decision tree out loud: Is this a repair, minor alteration, or major alteration? Does it change the type design? What data source applies? Show that you can identify the correct approval path—not just pick the right answer, but explain why.

STEP-BY-STEP PROCEDURE

1. Read the work order or customer request and identify exactly what is being done — removal, replacement, repair, or alteration. Be specific: 'replace a failed KX-155 comm radio with an identical refurbished KX-155' is a repair; 'replace a KX-155 comm with a Garmin GTN-650Xi' is an alteration.
2. Classify the work against 14 CFR 1.1 and Part 43 Appendix A. Ask: Does this appreciably affect weight, balance, structural strength, performance, powerplant operation, or flight characteristics? Does it change the type design? If yes to either, it is a major alteration.
3. If repair, classify against Appendix A(b): Is it one of the listed major repairs (control surface, pressure vessel, flight-critical wiring)? If yes, major repair. If no, minor repair.

4. Select the data tier: Tier 1 (manufacturer ICA/CMM) for repairs within type design; Tier 2 (STC) for major alterations; Tier 3 (DER 8110-3) for one-off engineering approvals; Tier 4 (AC 43.13-1B via field-approved 337) when no STC exists; Tier 5 (DOA/ODA) when OEM delegation covers the scope.
5. Verify aircraft eligibility: Cross-check the STC's Approved Model List against the aircraft's make, model, and serial number. AML is on page 1 or 2 of the STC package — if the aircraft is not on the list, the STC is not approved data for this installation.
6. Verify the data version is current: Pull the latest revision of the STC, CMM, or AC from the controlling source (FAA website, manufacturer portal, shop library). Outdated revisions may have been superseded for safety reasons.
7. Document the data selection in the work package: data source name, revision date, revision number, and the specific sections/drawings used. This becomes part of the audit trail for the 337 or logbook entry.
8. Before starting the work, brief the evaluator (or supervisor) on your selection and rationale. A confirmation step at this stage prevents a disqualifying data error downstream.
9. If in doubt, escalate: shop DER, shop ACSI (Avionics Certification Shop Inspector), or the FSDO. The cost of an escalation call is 15 minutes; the cost of a wrong data choice is rework, enforcement, or worse.

EVALUATOR IS WATCHING FOR

- Candidate explicitly classifies the work as major/minor alteration or major/minor repair, citing 14 CFR 1.1 or 43 Appendix A.
- Candidate names the specific data tier and why it applies (not just 'use the STC' — but 'STC because this is a major alteration and an STC exists covering this aircraft').
- AML cross-check performed against aircraft-on-file (make, model, S/N).
- Data revision current — not a cached PDF from three years ago.
- Candidate articulates what documentation product follows from the classification (337 for majors, 43.9 entry for minors).

COMMON MISTAKES

- Assuming 'the manufacturer's install manual is enough.' Manufacturer data authorizes the physical installation but does not replace the STC/337 requirement for a major alteration.
- Using an STC without verifying the AML. An STC for a Cessna 182 is not valid for a 182RG unless the RG variant is listed.
- Defaulting to AC 43.13-1B for any installation. AC 43.13-1B is acceptable data for minor alterations but requires FSDO field approval on a 337 to serve as approved data for majors.
- Failing to note the data revision in the work package. An auditor asks 'which revision of the STC did you use?' and the answer must be in the records.
- Skipping the AD check on the new equipment. ADs apply to the equipment being installed, not only to the aircraft.

Task 1.4.2 — AD Applicability Search

How to demonstrate: Using the FAA AD database or your shop's AD tracking system, perform a search for a specific avionics unit by part number and manufacturer. Document your search method (which database, what search terms, date of search) and results (ADs found or 'no applicable ADs found'). This should become a standard part of every work package.

STEP-BY-STEP PROCEDURE

1. Gather the complete identification for everything being worked on: aircraft make/model/serial, engine make/model/serial, propeller make/model/serial, and the part numbers of any appliance being installed, removed, or affected.
2. Open the FAA Dynamic Regulatory System (drs.faa.gov) — the authoritative AD source that replaced rgl.faa.gov. Alternatively use a commercial AD tracker (ADLog, CMI, CAMP) if the shop subscribes.
3. Search by aircraft model and serial range. Record the search filters exactly as entered — screenshot if possible for the work package.
4. Search by engine and propeller — ADs apply to these independently of airframe ADs.

5. Search by appliance part number for any avionics unit being installed or removed. Appliance ADs attach to the part number regardless of what aircraft it ends up on.
6. Review each AD returned: applicability block, effective date, compliance required-by date, and method of compliance. Capture these in a spreadsheet or AD list.
7. For recurring ADs, calculate the next compliance date using the aircraft's logbook data (hours, cycles, calendar time). Double-check the recurring interval — some ADs change intervals between revisions.
8. Document in the work package: search date, database used, search terms, and results (including 'none applicable' if none found — this is required evidence of due diligence, not optional).
9. If an AD is identified, plan the work: does your shop have the ops specs rating and the approved data to comply? If not, escalate to the owner/operator — they are responsible under 91.403(a) but cannot comply without a capable shop.

EVALUATOR IS WATCHING FOR

- Candidate goes to drs.faa.gov (or equivalent authoritative source) — not a random Google search.
- All three elements searched: airframe, engine/propeller, and appliance part numbers.
- Search date, database, and query terms recorded in the work package.
- Recurring AD next-compliance date calculated, not just 'AD is complied with.'
- 'None applicable' documented when no ADs are found — shows candidate understands the absence of ADs is itself a recordkeeping result.

COMMON MISTAKES

- Searching only the airframe and forgetting engine/prop/appliance. ADs attach at the product level, not the aircraft level.
- Using rgl.faa.gov from an old bookmark — it has been replaced by drs.faa.gov and may contain stale information.
- Failing to record the search, so a future auditor cannot reproduce the due diligence.
- Miscounting recurring compliance dates — especially when AD intervals include 'whichever comes first' (hours OR calendar time).
- Treating 'no ADs found' as no documentation needed. 91.417 requires a current AD status; the absence of ADs on a given date is itself a recordkeeping entry.

Task 1.4.3 — Major vs. Minor Alteration

How to demonstrate: Be able to clearly state: a major alteration appreciably affects weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other qualities affecting airworthiness (per Part 1). A minor alteration does not. Then give real examples from your shop.

STEP-BY-STEP PROCEDURE

1. Read the definition in 14 CFR 1.1: 'Major alteration means an alteration not listed in the aircraft, aircraft engine, or propeller specifications — (1) That might appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other qualities affecting airworthiness; or (2) That is not done according to accepted practices or cannot be done by elementary operations.'
2. Apply the first prong: Does the alteration appreciably affect weight/balance/structure/performance/powerplant/flight-characteristics? Think concretely — a new autopilot adds weight, changes CG, interacts with flight controls, yes to all four.
3. Apply the second prong: Can a competent technician accomplish the work by elementary operations using standard shop practices? If the work requires engineering analysis, load testing, or certified structural repair — it is a major alteration regardless of weight/balance.

4. Cross-reference Part 43 Appendix A(a) for the illustrative list: Airframe majors include changes to wings, tail surfaces, fuselage structure, fuel tanks, control systems, landing gear, retractable landing gear attachment. Powerplant majors include conversions, supercharger installations, changes to accessory drives. Appliance majors include changes that affect operation, ratings, or limitations.
5. If the alteration is installation of a new avionics unit: is the unit new to the aircraft (not a like-for-like)? Does it interface with flight instruments, autopilot, or flight-critical wiring? Does it need new antennas drilled? Does it draw power from a protected bus? Yes to any — treat as major.
6. If minor: document with a 14 CFR 43.9 logbook entry citing the data source, the work performed, and the return to service statement. No 337.
7. If major: plan for a Form 337 (43 Appendix B), cite the STC or other Approved Data in Block 3, describe the work in Block 8, and file the 337 within 48 hours of RTS per Appendix B(a)(3).
8. When you are not sure, treat the work as major. The cost of overdocumenting is filling out a 337 that was not required. The cost of underdocumenting is an enforcement action.

EVALUATOR IS WATCHING FOR

- Candidate cites 14 CFR 1.1 and/or 43 Appendix A, not 'I think it's a major because...'
- Candidate applies both prongs of the major-alteration test (the appreciable-affect prong AND the elementary-operations prong).
- Candidate can give at least two concrete examples from their own recent work, not only textbook scenarios.
- Candidate knows the documentation product: 337 for major, 43.9 for minor.
- Candidate errs on the side of major when the case is close — this is a professional-judgment red flag if they do not.

COMMON MISTAKES

- Using 'small in physical size' as the test for minor. A 1-ounce GPS antenna that drives an autopilot is a major alteration; a 30-pound radio that replaces an identical 30-pound radio is a repair.
- Calling an avionics upgrade 'just a swap' when the replacement has different capabilities, interfaces, or certifications.
- Thinking the manufacturer calls the shot. The technician and the FAA classify alterations per 14 CFR 1.1 and 43 Appendix A — the manufacturer's marketing material does not.
- Forgetting that 'performance' in 1.1 includes fuel burn, range, cruise speed, ceiling — not only takeoff performance.
- Forgetting the 48-hour 337 submission deadline once you correctly classify the work.

Task 1.4.4 — GPS Navigator Installation Approvals

How to demonstrate: Walk through the complete approval chain: (1) STC required because this is a major alteration (new system, new antennas, new wiring). (2) STC specifies approved aircraft list—verify the customer’s aircraft is on it. (3) Follow the STC installation instructions exactly. (4) Complete Form 337 documenting the major alteration. (5) Logbook entry referencing the STC. (6) AD check on the new equipment. (7) Weight and balance update if required by STC.

STEP-BY-STEP PROCEDURE

1. Confirm the scope is a major alteration under 14 CFR 1.1 and 43 Appendix A(a): new IFR GPS navigator, new antenna(s), new wiring, interface with existing nav/com/autopilot. This is unambiguously a major alteration.
2. Acquire the STC package from the navigator manufacturer (Garmin, Avidyne, etc.). Verify the STC approval letter, the AML, the installation drawings, the master drawing list, the Flight Manual Supplement (AFMS), and the ICA. All must be current revision.
3. Cross-check the AML: aircraft make, model, serial number, engine type (if the AML is engine-dependent). If the aircraft is NOT on the AML — stop. The STC cannot be

used. Options: find a different STC that covers this airframe, pursue a field approval, or refuse the work.

4. Run AD searches: against the aircraft, the navigator part number, any new antennas, and any interfaced components (e.g., transponder, autopilot). Document results in the work package.

5. Install exactly per the STC installation drawings: antenna location and clearances, wire routing (separation from pitot-static, fuel lines, engine controls), connector pinouts, shield terminations, structural mounting. Deviation from the drawings voids the STC approval.

6. Install the AFMS in the aircraft's AFM per the STC requirements. Some STCs require an AFMS Supplement to be physically inserted; others require pilots to reference it.

7. Execute the functional checkout per the STC's checkout procedure. Typical elements: self-test pass, antenna patterns verified, database current, CDI deflection interface correct, autopilot coupled GPS mode functional.

8. Complete FAA Form 337: Block 1 identifies the aircraft; Block 3 identifies the Approved Data (STC number, revision, AML eligibility); Block 8 describes the work; Block 6 is the return-to-service signature. Attach a copy of the AFMS and a copy of the STC approval letter to the 337 file.

9. Within 48 hours of aircraft return to service, submit one copy of the 337 to FAA AFS-620 Aircraft Registration Branch per 43 Appendix B(a)(3). The owner's copy is retained with aircraft records per 91.417(b)(2). Update the weight-and-balance record if the STC changes empty weight or CG — attach the revised W&B to the 337 package.

EVALUATOR IS WATCHING FOR

- AML cross-check evidence in the work package (not just 'I checked').
- STC revision is current and referenced in Block 3 with exact certificate number and revision date.
- AD checks performed on the navigator part number, not only the airframe.
- Functional checkout signed off with test equipment calibration dates where applicable.
- Form 337 complete, accurate, and ready for submission within 48 hours of RTS.
- AFMS physically in the aircraft (if required) and noted on the 337.

COMMON MISTAKES

- Installing the box per the manufacturer's install manual without the STC AML cross-check. The install manual describes HOW; the STC authorizes WHETHER.
- Treating the 337 as 'paperwork to do later.' The 48-hour submission deadline is a regulatory requirement, not an internal shop preference.
- Skipping the W&B update because 'it's only a couple of pounds.' Any STC that changes weight requires a revised W&B under 91.9 and the STC's ICA.
- Routing the new wiring next to pitot-static or fuel lines because 'that's where the space is.' Route per the STC drawings — separation requirements are written for reasons.
- Forgetting to verify the GPS database currency at turnover. A new GPS with an expired database is not IFR-usable and customer dissatisfaction follows immediately.

PART IV — BACKGROUND THEORY

The Science Behind the Procedures

When the oral board examiner asks *why*, these are the explanations they expect to hear.

WHY THE 337 IS A LEGAL DOCUMENT NOT PAPERWORK

FAA Form 337 is not a form you fill out; it is a sworn statement that carries the force of federal law. Block 3 is the Approved Data certification — the signer is certifying under penalty of 14 CFR 43.12 (falsification) and 18 USC 1001 (false statements to federal agency) that the data used is in fact approved for this specific aircraft and alteration. Block 6 is the Return to Service signature — the signer is certifying under 14 CFR 43.7 that the aircraft is airworthy after the described work. Block 8 describes the work, and the narrative must match the Approved Data cited in Block 3 and the physical installation on the aircraft. When a 337 is filed with the FAA Aircraft Registration Branch, it enters the permanent aircraft record — it will follow that aircraft for the rest of its service life, transfer with title on sale, and be reviewed by every future buyer's pre-purchase inspector and by every IA performing annuals. Twenty years from now, the 337 you signed today will be read by someone else, and your name is on it. This is why 337s are treated as evidence in enforcement actions — a falsified or materially incomplete 337 is a documented paper trail of the violation. FSDO inspectors routinely cross-check 337s against the physical aircraft during surveillance, and discrepancies between Block 8 narrative and observable installation drive certificate actions under 49 USC 44709. The 48-hour filing deadline in 43 Appendix B(a)(3) exists so that the owner can legally operate the aircraft (the 337 is proof the alteration is approved) and so the FAA's records are current for the next inspection. Late or missing 337s ground aircraft and trigger audits.

WHY STC VS TC MATTERS FOR LIABILITY

The Type Certificate defines the aircraft as designed and produced by the OEM — the liability for airworthiness of the TC configuration rests with the TC holder (e.g., Textron for a Cessna 172). When you install an STC-approved alteration, you are modifying the TC configuration, and the liability chain restructures. The STC holder (e.g., Garmin for a GTN 650Xi STC) becomes responsible for the airworthiness of the altered configuration, but only for the alteration itself — the interface between the STC-installed equipment and the rest of the airframe, the installation workmanship, and the ongoing airworthiness of the modification all become the responsibility of the installing technician, the repair station, and ultimately the owner/operator under 91.403. This is why STC ICAs matter: they are the STC holder's written maintenance procedures for the altered configuration, and compliance with the ICA is mandatory for airworthiness. When a post-STC failure causes damage, the post-accident investigation will parse the liability: Did the STC itself have a defect (STC holder liability)? Was the installation done per the STC drawings (installer liability)? Were the ICA procedures followed in ongoing maintenance (owner/operator and subsequent technician liability)? A technician who installs an STC but deviates from the drawings assumes liability for the deviation — the STC is no longer covering the work, and the installer has effectively performed an unapproved alteration. This is also why a field-approved 337 without an STC is considered higher-risk: the DER or FSDO inspector approving the one-off data accepts responsibility for the data's adequacy for that single aircraft, but the installing technician still owns workmanship and ongoing airworthiness. The STC model exists precisely to distribute this liability across capable parties; bypassing it (installing without approval, falsifying Block 3, or deviating from drawings) collapses the entire liability chain onto the technician and the repair station.

WHY MEL DEFERRAL REQUIRES DOCUMENTED MAINTENANCE ACTION

The MEL (Minimum Equipment List) exists to permit dispatch of an aircraft with specified inoperative equipment under controlled conditions. Under 14 CFR 91.213(a), an MEL operation is only legal when the MEL has been approved by the operator's FSDO (via a Letter of Authorization for Part 91, or as part of OpSpecs for Part 121/135) and the deferral is executed per the MEL's procedures. Every MEL item has an associated 'M' procedure (maintenance action) and 'O' procedure (operational procedure). The M procedure is not optional — it is a required maintenance action that must be performed before the aircraft can be dispatched with the item inoperative. Examples: if the #2 VHF comm is MEL-deferred, the M procedure typically requires the circuit breaker to be pulled and collared and the unit to be placarded 'INOP'; the O procedure typically requires the PIC to verify alternate communication capability. The legal basis: the MEL is an alteration to the operating limitations of the aircraft for the duration of the deferral, and the M procedure ensures the aircraft is still airworthy within the reduced-capability envelope. Without the M procedure, the aircraft is not airworthy under 91.7 regardless of what the MEL says. The maintenance action must be documented in the aircraft logbook under 14 CFR 43.9 — not just a sticker in the cockpit. The entry must identify the MEL item, the date deferred, the compliance date by which repair must occur (typically Category A, B, C, or D — from immediate to 120 days), the M procedure accomplished, and the technician's signature. Failure to document the M procedure turns an authorized MEL dispatch into an unauthorized flight with inoperative equipment — a direct 91.213 and 91.405 violation against the operator, and a 43.9 recordkeeping violation against the technician. The MEL is the most misunderstood regulation in the fleet because pilots treat it as 'permission to fly broken' and technicians sometimes treat it as 'someone else's problem.' In reality, the MEL is a controlled-risk tool that only works when both the M procedure and the documentation are executed exactly as approved.

PART V — SELF-CHECK

Oral Board Practice Questions

Answer each aloud before opening the answer key. Mastery means you can explain it to a non-technical person, not just recite a definition.

Questions

1. Q1: A customer brings you a Cessna 172 and wants you to install a Garmin GTN 650Xi to replace their old King KX 155. Walk me through the approved data you need and why.
2. Q2: What is the difference between maintenance and an alteration? Give me an avionics example of each.
3. Q3: Your shop has a Part 145 certificate with limited radio ratings. A customer asks you to install a new autopilot. Can you do it? How do you determine that?
4. Q4: You are about to install a new comm antenna. Where and how do you check for applicable ADs?
5. Q5: A technician in your shop says, 'AC 43.13-1B covers any antenna installation.' Is that correct? When does AC 43.13-1B apply and when does it not?
6. Q6: Explain what 14 CFR 91.403 says about maintenance responsibility. Who is ultimately responsible for the airworthiness of the aircraft—the technician, the repair station, or the owner?

Answer Key

▶ **Q1: A customer brings you a Cessna 172 and wants you to install a Garmin GTN 650Xi to replace their old King KX 155. Walk me through the approved data you need and why.**

▶ **Q2: What is the difference between maintenance and an alteration? Give me an avionics example of each.**

▶ **Q3: Your shop has a Part 145 certificate with limited radio ratings. A customer asks you to install a new autopilot. Can you do it? How do you determine that?**

▶ **Q4: You are about to install a new comm antenna. Where and how do you check for applicable ADs?**

▶ **Q5: A technician in your shop says, 'AC 43.13-1B covers any antenna installation.' Is that correct? When does AC 43.13-1B apply and when does it not?**

▶ **Q6: Explain what 14 CFR 91.403 says about maintenance responsibility. Who is ultimately responsible for the airworthiness of the aircraft—the technician, the repair station, or the owner?**

Written Exam Crosswalk

The knowledge in this section maps to Module 13 (Documentation) on the written exam, specifically Q161–165. Key exam topics from this section: When a logbook entry vs. Form 337 is required Who can sign a return-to-service statement The distinction between maintenance and alteration Documentation requirements for like-for-like replacements vs. new installations Approved data hierarchy and when STCs are mandatory Study these topics in connection with PQS Section 2 (Maintenance Documentation), which covers the hands-on documentation skills tested on the exam. End of Section 1 — Regulatory Compliance & Approved Data

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